

Create the C program using fork()

```
saireddy@DESKTOP-HKQRL9G:~/session$ ./process
-bash: ./process: No such file or directory
saireddy@DESKTOP-HKQRL9G:~/session$ gcc session.c -o session
cc1: fatal error: session.c: No such file or directory
compilation terminated.
saireddy@DESKTOP-HKQRL9G:~/session$ mkdir session
saireddy@DESKTOP-HKQRL9G:~/session$ cd session
saireddy@DESKTOP-HKQRL9G:~/session/session$ nano process.c
saireddy@DESKTOP-HKQRL9G:~/session/session$ gcc process.c -o process
saireddy@DESKTOP-HKQRL9G:~/session/session$ ./process
Parent Process Started
Parent PID : 602
Parent PPID : 367

--- Parent Process ---
Parent PID : 602
Child PID : 603
Parent is waiting for child...

--- Child Process ---
Child PID : 603
Child PPID : 602
Child is running...
Child process terminated.
Child completed.
Parent process terminated.
saireddy@DESKTOP-HKQRL9G:~/session/session$ |
```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
#include <stdlib.h>
```

```
int main() {
```

```
    pid_t pid;
```

```
    printf("Parent Process Started\n");
```

```
    printf("Parent PID : %d\n", getpid());
```

```
    printf("Parent PPID : %d\n", getppid());
```

```
    pid = fork();
```

```
    if (pid < 0) {
```

```
        perror("fork failed");
```

```
        return 1;
```

```
    }
```

```
if (pid == 0) {  
    // Child process  
    printf("\n--- Child Process ---\n");  
    printf("Child PID : %d\n", getpid());  
    printf("Child PPID : %d\n", getppid());  
  
    printf("Child is running...\n");  
    sleep(10);  
  
    printf("Child process terminated.\n");  
    exit(0);  
}  
else {  
    // Parent process  
    printf("\n--- Parent Process ---\n");  
    printf("Parent PID : %d\n", getpid());  
    printf("Child PID : %d\n", pid);  
  
    printf("Parent is waiting for child...\n");  
    wait(NULL);  
  
    printf("Child completed.\n");  
    printf("Parent process terminated.\n");  
}  
  
return 0;  
}
```

```

saireddy@DESKTOP-HKQRL9G: ~/session/session$ gcc fork.c -o fork
cc1: fatal error: fork.c: No such file or directory
compilation terminated.
saireddy@DESKTOP-HKQRL9G:~/session/session$ ./fork
-bash: ./fork: No such file or directory
saireddy@DESKTOP-HKQRL9G:~/session/session$ nano fork.c
saireddy@DESKTOP-HKQRL9G:~/session/session$ gcc fork.c -o fork
saireddy@DESKTOP-HKQRL9G:~/session/session$ ./fork
Parent Process
Parent PID : 640
Child PID : 641
Parent is waiting for child...
Child Process
Child PID : 641
Child PPID : 640
Child is running...
Child process terminated.
Child completed.
Parent process terminated.
saireddy@DESKTOP-HKQRL9G:~/session/session$

```

```
#include <stdio.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
#include <stdlib.h>
```

```
int main()
```

```
{
```

```
    pid_t pid = fork();
```

```
    if (pid < 0)
```

```
    {
```

```
        printf("Fork failed\n");
```

```
    }
```

```
    else if (pid == 0)
```

```
    {
```

```
        printf("Child Process\n");
```

```
        printf("Child PID : %d\n", getpid());
```

```
        printf("Child PPID : %d\n", getppid());
```

```
        printf("Child is running...\n");
```

```
        sleep(5);
```

```
    printf("Child process terminated.\n");
}
else
{
    printf("Parent Process\n");
    printf("Parent PID : %d\n", getpid());
    printf("Child PID : %d\n", pid);

    printf("Parent is waiting for child...\n");
    wait(NULL);

    printf("Child completed.\n");
    printf("Parent process terminated.\n");
}

return 0;
}
```